

CALIBRATION OF ORIFICE METER

ChE 206: Chemical Engineering Laboratory I

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BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

ChE 206: Chemical Engineering Laboratory I

Experiment No 05: CALIBRATION OF ORIFICE METER

Submitted To:

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Statement of Academic Integrity

By signing this experiment, I hereby declare that the work on the experiment titled "Calibration of Orifice Meter" has been prepared honestly and responsibly based on the experimental work, observations, calculations and analysis carried out during the laboratory session. All information, equations, figures, and data obtained from books or other external sources have been properly acknowledged and cited. No data have been fabricated, altered, or intentionally misrepresented, and the interpretations presented in this report reflect the actual experimental findings to the best of my knowledge. I accept full responsibility for the accuracy and originality of the work submitted.

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Signature

Summary

The primary objective of this experiment was to calibrate a sharp-edged concentric orifice meter by evaluating its discharge coefficient across various flow rates. The operation of the device is grounded in the continuity equation and Bernoulli's principle, where a localized cross-sectional restriction accelerates the flowing fluid, converting static pressure energy into kinetic energy. The resulting differential pressure head (Δh) was measured across the upstream and downstream pressure taps of a horizontal pipe using a U-tube differential manometer filled with carbon tetrachloride (CCl_4), while clean water was circulated as the working fluid by a centrifugal pump. Because theoretical models assume frictionless flow and neglect energy dissipation as well as jet contraction at the vena contracta, the coefficient of orifice (C_d) is introduced to reconcile the actual discharge with ideal predictions. During testing, the experimental volumetric flow rate (Q) was varied from Q_{min} to Q_{max} , producing a corresponding pressure head difference (Δh) between h_{min} and h_{max} . The calculated Reynolds numbers (Re) ranged from Re_{min} to Re_{max} , confirming that fully developed turbulent flow ($Re > 2300$) was sustained throughout the investigation. As predicted by fluid dynamic theory, the experimental flow rate was consistently lower than the ideal theoretical discharge due to viscous shear, wall friction, turbulent eddy dissipation, and flow separation. From the graphical analysis, a strong linear calibration trendline ($Q \propto \sqrt{\Delta h}$) was established between Q and $\sqrt{\Delta h}$, validating the theoretical proportionality derived from Bernoulli's equation. Furthermore, the plots of C_d versus Re , Q , and Δh exhibited nearly horizontal trendlines, confirming that the discharge coefficient remains essentially constant in the fully turbulent flow regime. However, the calculated average value of C_d was found to be moderately higher than the standard theoretical range (0.61 ± 0.02) for sharp-edged orifices. This upward deviation and minor data scatter were caused by experimental uncertainties, including pressure fluctuations from the pump, and gradual edge wear or rounding on the upstream orifice face, which reduced jet contraction and yielded a larger effective vena contracta area. In conclusion, the orifice meter was successfully calibrated, demonstrating that once an empirical calibration curve is established, it can be reliably used for direct and accurate flow measurement in industrial pipelines.

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Table of Contents

Statement of Academic Integrity i

Summary ii

Acknowledgement iii

Table of Contents iv

Table of Illustration v

1. Introduction 1

2. Theory 2

2.1 Orifice Meter & Its Working Principle 2

2.2 Vena Contracta 2

2.3 Continuity Equation 3

2.4 Bernoulli's Equation 3

2.5 Differential Manometer Relation 3

2.6 Reynolds Number 4

2.7 Types of Orifice Plates 4

2.8 Coefficient of Orifice 5

3. Experimental Section 6

3.1 Apparatus and Materials 6

3.1.1 Apparatus 6

3.1.2 Fluids 6

3.2 Experimental Setup 6

3.3 Experimental Procedure 7

4. Results & Discussion 8

4.1 Calibration Results of the Orifice Meter 8

4.2 Correlation Between Experimental and Theoretical Values 10

4.3 Experimental Errors and Their Consequences 11

5. Conclusion 12

References 13

Nomenclature 14

Appendices 15

Appendix A: Observed Data 15

Appendix B: Calculated Data 16

Appendix C: Sample Calculation (Observation No. 3) 17

Appendix D: Derivation of Theoretical Orifice Equation 18

Table of Illustrations

List of Figures

Figure 1: Schematic diagram of an orifice meter 2

Figure 2: Different types of orifice plates based on geometry 4

Figure 3: Schematic diagram of orifice meter apparatus 6

Figure 4: Plot of Pressure head difference vs volumetric flow rate 8

Figure 5: Plot of square root of pressure head difference vs volumetric flow rate 9

Figure 6: Plot of coefficient of discharge vs Reynolds number 9

Figure 7: Plot of coefficient of orifice vs pressure head difference 10

Figure 8: Plot of coefficient of orifice vs square root of pressure head difference 10

List of Tables

Table 1: Data for volumetric flow rate, pressure head difference, coefficient of orifice, and Reynolds number 8

Table 2: Observed data for volume, time of collection, and manometer reading 15

Table 3: Calculated data for experimental volumetric flow rate, pressure head difference, and square root of pressure head difference 16

Table 4: Calculated data for average velocity, theoretical volumetric flow rate, coefficient of orifice, and Reynolds number 16

1. Introduction

Fluid flow measurement is a fundamental requirement across chemical processing plants, oil refineries, water supply networks, and power generation facilities. It is performed to monitor industrial processes, ensure plant safety, maintain product quality, and account for material balances during production. Flow rates are measured by placing specialized instruments directly in pipelines to determine the volume or mass of fluid passing through a given cross-section per unit time. Among various flow-measuring tools, differential pressure flow meters, particularly orifice meters are widely chosen due to their low cost, simple construction, lack of moving parts, and reliable performance.

An orifice meter operates by inserting a flat plate with a concentric circular opening perpendicular to the fluid flow. As fluid passes through this restricted opening, its cross-sectional area decreases and its velocity increases. According to the continuity equation and Bernoulli's principle, this acceleration causes a drop in static pressure immediately downstream of the plate. The streamlines continue to converge past the physical opening, reaching a minimum cross-sectional area and maximum velocity at a location known as the vena contracta. The resulting static pressure difference between the upstream pipe section and the downstream restriction is measured using a differential U-tube manometer filled with a high-density liquid like carbon tetrachloride.

Ideal flow equations derived from Bernoulli's principle overestimate actual volumetric discharge because real fluids experience frictional dissipation, viscous drag, wall turbulence, and flow separation. To adjust for these mechanical energy losses, an empirical factor known as the coefficient of orifice is introduced into the theoretical model.

Subsequent flow rates can be calculated directly from pressure head readings without collecting fluid manually, as long as density and viscosity changes are included in the Reynolds number. However, experimental difficulties are frequently met in practice, such as pressure variations caused by pumps and slight rounding of the orifice edge, both of which disrupt reading accuracy and change the discharge coefficient.

2. Theory:

2.1 Orifice Meter & It's Working Principle

An orifice meter is a differential pressure type flow measuring device used to determine the volumetric flow rate of a fluid flowing through a pipeline. It consists of a thin plate with a sharp-edged circular opening, called an orifice, which is installed perpendicular to the direction of flow. A schematic diagram of a typical orifice meter is shown in Figure 1[1]

When fluid flows through the pipe and passes through the orifice, the effective flow area suddenly reduces. Due to this reduction in area, the velocity of the fluid increases. According to Bernoulli's principle, an increase in velocity is accompanied by a decrease in static pressure. As a result, a pressure difference is created between the upstream section and the downstream section of the orifice. This pressure difference is measured using a differential manometer and is directly related to the flow rate of the fluid. Thus, by measuring the pressure drop across the orifice, the volumetric flow rate can be determined.

The working principle of the orifice meter is therefore based on the conversion of pressure energy into kinetic energy, as described by the continuity equation and Bernoulli's equation.[2]

2.2 Vena Contracta

When fluid flows toward and passes through a sharp-edged orifice, fluid streamlines cannot execute abrupt turns around the sharp metal edges due to fluid inertia. Instead, the converging streamlines continue to contract even after exiting the orifice opening, forming a constricted fluid jet. The location downstream where the cross-sectional area of the fluid jet reaches its absolute minimum is defined as the Vena Contracta. Downstream of the vena contracta, the fluid jet expands to refill the full pipe diameter. This sudden boundary expansion creates severe flow separation, wall friction, and recirculation eddies. These turbulent disturbances convert mechanical energy into unrecoverable thermal energy, causing a permanent head loss across the meter. This structural energy dissipation requires introducing the discharge coefficient during experimental calibration.

2.3 Continuity Equation

For steady, incompressible flow, the continuity equation is given by

where, A = Cross-sectional area v = Average velocity

In an orifice meter, the flow area reduces at the orifice opening, and according to the continuity equation, the fluid velocity increases at this section. This increase in velocity leads to a corresponding decrease in static pressure. Thus, the continuity equation explains the rise in velocity that contributes to the pressure difference used for flow measurement.

2.4 Bernoulli's Equation

For steady, incompressible, and frictionless flow between two sections along a streamline, Bernoulli's equation is expressed as

For a horizontal pipe,

Hence,

Here, P_1 and P_2 represent the static pressures at sections 1 and 2, respectively. For a horizontal pipe, the elevation terms (z_1 & z_2) cancel out, and the pressure difference is balanced by the change in velocity head. This relation forms the theoretical basis for pressure-based flow measurement devices such as the orifice meter.

2.5 Differential Manometer Relation

The pressure difference across the orifice is measured using a U-tube manometer containing carbon tetrachloride (CCl_4). The equivalent pressure head in terms of water is given by

where, R = manometer reading difference = density of manometer fluid = density of water

In an orifice meter, the sudden reduction in flow area causes an increase in velocity and a corresponding drop in static pressure across the orifice. This pressure difference between the upstream and downstream sections is measured using a differential manometer. The manometer reading is then converted into an equivalent pressure head in terms of water, which is directly used to determine the discharge through the orifice.

2.6 Reynold's Number

The Reynolds Number (Re) is a dimensionless parameter in fluid mechanics representing the ratio of inertial forces to viscous forces within a moving fluid.

where, ρ = density of fluid v = average velocity D = pipe diameter μ = dynamic viscosity of fluid

In the context of orifice meter calibration, the Reynolds number determines the stability of the discharge coefficient (C_d). In laminar or transitional flow regimes, C_d varies significantly with changes in velocity because boundary layer thickness changes with flow rate. However, in fully turbulent flow ($Re > 10^4$), inertia forces dominate over viscous forces, stabilizing jet contraction at the vena contracta. As a result, C_d becomes nearly constant and independent of Reynolds number variations.

2.7 Types of Orifice Plates

Based on the geometry of the orifice plate, several types are used depending on the nature of the flowing fluid. The sharp-edge orifice plate is the most commonly used type for clean fluids and turbulent flow conditions. The eccentric orifice plate has an off-center opening and is suitable for fluids containing suspended solids, as it prevents sediment accumulation. The segmental orifice plate has a segment-shaped opening and is used when the flow contains gas bubbles or solid particles. The conical orifice plate has a conical inlet profile and provides smoother flow entry with reduced energy loss.[3]

2.8 Coefficient of Orifice

The theoretical discharge equation assumes ideal conditions with no energy losses. However, in real flow, losses occur due to viscosity, turbulence, flow separation, and vena contracta formation. Therefore, the actual discharge is always less than the theoretical discharge.

To account for these effects, the coefficient of orifice is introduced:

Hence,

For an ideal fluid, $C_d = 1$, but in practice its value is always less than unity. Because, when real fluid flows through an orifice, energy losses occur due to turbulence and flow separation, which cause the actual discharge to be lower than the theoretical discharge predicted by Bernoulli's equation

3. Experimental Section

3.1 Apparatus and Materials

3.1.1 Apparatus

Orifice plate

Differential U-tube manometer

Flowmeter

Centrifugal pump

Reservoir

Pressure control valve

Pipe fittings (elbows, joints, connectors)

Scale attached with manometer

Thermometer

Stopwatch

3.1.2 Fluids

Water

Carbon tetrachloride

3.2 Experimental Setup

The experimental setup consists of a concentric orifice meter for flow measurement, a centrifugal pump for fluid circulation, a differential U-tube manometer for recording pressure head differences, and a flow meter to measure discharge. Water is pumped from a storage reservoir through a pipeline containing the orifice plate, where an inlet bypass line is installed to regulate the flow rate and prevent reservoir overflow. The entrance valve is used to establish the maximum flow limit, while the outlet control valve is adjusted to stabilize and set specific flow rates. Pressure tapping points located upstream and downstream of the orifice plate are connected directly to the differential U-tube manometer filled with carbon tetrachloride (CCl_4) as the indicating liquid

3.3 Experimental Procedure:

Initially, the supply valve was opened to fill the storage reservoir with water to provide a continuous fluid feed to the centrifugal pump. The entrance valve and the bypass valve were then adjusted to establish a steady, uniform flow rate while defining the upper flow limit and preventing tank overflow. Prior to recording measurements, the differential U-tube manometer was inspected to ensure that both fluid limbs were balanced at the same initial datum level. Once secure pressure tap connections were confirmed, the control

valve was adjusted to set the initial flow rate, allowing sufficient time for the flow to stabilize before measurements were taken. To determine the experimental discharge, a fixed volume of water corresponding to half a cycle of the flowmeter () was collected, and the time interval was measured using a stopwatch. Under identical operating conditions, the liquid column heights in the left and right limbs of the manometer were recorded to evaluate the differential pressure head across the orifice plate. This measurement sequence was repeated for seven additional runs at incrementally higher flow rates. Finally, both the room temperature and water temperature were measured using a thermometer to determine the physical properties (density and viscosity) of the carbon tetrachloride and water.

4. Result & Discussion

The primary objective of this experiment is to calibrate a sharp-edge concentric orifice meter to evaluate empirical relationship between the differential pressure head and the volumetric flow rate using water as the working fluid and carbon tetrachloride as the manometer fluid.

4.1 Calibration Results of the Orifice Meter

Table 1: Data for volumetric flow rate, pressure head difference, coefficient of orifice and reynold's number

The experimental results obtained from the calibration of the orifice meter show that as the experimental volumetric flow rate was increased from to , the corresponding pressure head difference () rose steadily from to . This behavior agrees with the principle derived from Bernoulli's theorem and the continuity equation, where fluid acceleration through the restricted orifice causes a localized drop in static pressure that scales quadratically with flow rate (), resulting in a linear increase in with discharge. The calculated Reynolds numbers () ranged from to , confirming that fully developed turbulent flow () was maintained throughout testing, which stabilizes the fluid jet at the vena contracta and ensures a nearly constant discharge coefficient. The experimental coefficient of orifice () was determined to lie between and with an average value of , adhering to the theoretical requirement of due to frictional losses, turbulence, and jet contraction. These deviations were primarily caused by experimental factors, including slight edge wear or rounding on the upstream face of the orifice plate which reduced streamline contraction and enlarged the effective vena contracta area. Overall, the experimental data demonstrated strong consistency with theoretical principles, confirming that the orifice meter was successfully calibrated for practical flow measurement.

From the experimental plot of differential pressure head vs volumetric flow rate, a distinct upward-curving trendline with a strong correlation coefficient of was obtained. This behavior is fully explained by Bernoulli's principle and the conservation of energy, where fluid acceleration through the restricted orifice causes a localized increase in kinetic energy at the direct expense of static pressure head. Because fluid velocity is directly proportional to volumetric flow rate, the induced static pressure drop theoretically varies with the square of the discharge (), predicting an ideal parabolic curve passing through the origin. The fitted experimental trendline closely matches this theoretical quadratic profile, confirming that the pressure drop increases progressively faster as higher flow rates are introduced.

From the experimental plot of the square root of differential pressure head versus volumetric flow rate, a strong linear calibration trendline with a correlation coefficient of was obtained. This linear response is directly supported by the governing orifice flow equation derived from Bernoulli's principle and the continuity equation, which establishes that the volumetric flow rate varies in direct proportion to the square root of the differential pressure head (). Theoretically, an ideal frictionless system should yield a perfectly straight line passing through the origin. The experimental data closely align with this predicted linear behavior, demonstrating that the orifice meter responds predictably and consistently to variations in pressure drop across the operating range, thereby confirming the suitability of this graph as a direct calibration standard.

Figure 6: Plot of coefficient of discharge vs reynolds number

From the experimental plot of the coefficient of discharge versus Reynolds number, the data points are observed to align along a nearly flat, horizontal trendline across the investigated Reynolds number range of to . This behavior is fully consistent with fluid dynamic theory for high Reynolds number regimes (), where inertial forces overwhelmingly dominate over viscous forces. In this fully turbulent zone, the hydrodynamic boundary layer remains thin and the streamline contraction pattern at the vena contracta stabilizes, causing the discharge coefficient to become virtually independent of the Reynolds number. Theoretically, a sharp-edged concentric orifice meter should exhibit a constant within the standard theoretical range of to . While the experimental trend correctly maintained the expected horizontal invariance, the measured values remained clustered around an average of approximately , showing minor scatter between and . This upward shift in and the slight data dispersion were primarily caused by experimental uncertainties, including slight edge wear or rounding on the upstream face of the orifice plate which reduced flow contraction and widened the effective jet area.

Figure 7: Plot of coefficient of orifice vs pressure head difference

From the experimental plot of the coefficient of discharge versus pressure head difference, the data points display a nearly flat profile with a very slight negative downward slope across the investigated range of to . Theoretically, because the coefficient of orifice is defined as the ratio of actual flow rate to theoretical flow rate () and both quantities scale proportionally with , the ratio is expected to remain invariant, predicting a perfectly horizontal line independent of the applied differential head. The observed minor downward trend occurs because higher differential heads induce substantially greater fluid velocities through the orifice restriction. As velocity increases, frictional shearing, turbulent eddy generation, and downstream flow separation dissipate progressively more kinetic

energy. Since theoretical equations assume frictionless flow, these intensified real-fluid losses cause the actual discharge to lag slightly behind theoretical predictions, resulting in a marginal decrease in .

Figure 8: Plot of coefficient of orifice vs square root of pressure head difference

The plot of the coefficient of orifice versus the square root of pressure head difference is theoretically expected to be horizontal, as the discharge coefficient should remain independent of the applied differential head. However, a slight negative slope is observed in the experimental plot because, at higher pressure differences, increased turbulent energy losses cause the theoretical flow rate to increase faster than the actual discharge rate, leading to a minor reduction in the calculated value.

4.2 Correlation between experimental and theoretical value

The experimental discharge was consistently lower than the theoretical discharge. This reduction is represented by the discharge coefficient being less than unity. The deviation between ideal theoretical predictions and actual experimental can be caused by two main fluid phenomena:

Fluid Contraction: The cross-sectional area of the fluid jet contracts to a minimum smaller area than the physical orifice opening downstream of the plate.

Energy Dissipation: Mechanical energy is converted into heat due to fluid viscosity, eddy formation, and sudden flow expansion downstream.

While standard theoretical values for sharp-edged orifice meters typically range from to , the experimental values obtained in this test averaged . This shift reflects specific mechanical and operational characteristics of the experimental apparatus as the orifice meter is using for long time.

4.3 Experimental Errors and Their Consequences

During the experiment, several sources of experimental error were identified, which influenced the accuracy and consistency of the obtained results. The most significant of error was associated with the differential manometer. As the accuracy of the scale is 0.5cm, we can't measure more accurate than it.

Another important source of uncertainty can be caused by the shape and geometry of the orifice plate. Slight rounding or wear on the upstream edge of the orifice plate reduces jet contraction. A reduced contraction increases actual flow area at the vena contracta, causing the measured values to be higher than standard sharp-edged theoretical values.

5. Conclusion

In this experiment, the performance of an orifice meter as a flow-measuring device was studied by comparing the theoretical discharge with the experimentally measured discharge obtained from volumetric measurements. The pressure head difference across the orifice was found to increase with an increase in flow rate, which is consistent with the theoretical relationship derived from Bernoulli's equation and the continuity principle. From the experimental data, the coefficient of orifice was evaluated and observed to remain approximately constant over the turbulent flow region, indicating that the orifice meter provides reliable results under such conditions.

The experimental discharge was generally lower than the theoretical discharge due to energy losses caused by friction, turbulence, and flow separation near the orifice plate. These deviations were accounted for by introducing the coefficient of orifice, which corrects the ideal flow equation for real fluid behavior. Minor discrepancies between theoretical and experimental values can be attributed to experimental uncertainties such as manometer reading errors, air bubbles in the pipeline, and limitations of the measuring instruments.

Overall, the experiment successfully demonstrated the working principle of an orifice meter and verified its suitability for measuring flow rate in pipelines, particularly in the turbulent flow regime. The results confirm that, after proper calibration, the orifice meter can be used as a practical and economical device for flow measurement in engineering applications.

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Nomenclature

Appendices

Appendix A

Observed Data:

Orifice diameter = 1inch

Pipe diameter = 2inch; Schedule No. 40

Room temperature = 29.5

Water temperature = 28

Table 2: Observed data for volume, time of collection, and manometer reading

Appendix B

Calculated Data:

Density of water at 28, = 996.24 kg/m³[4]

Viscosity of water at 28, = 0.0083238 Pa.s[4]

Density of CCl₄ at 29.5, = 1575.71 kg/m³[5]

Orifice diameter, D₀ = 1 inch = 0.0254 m

Pipe diameter (Schedule no. 40), D = 2.067 inch = 0.0525 m

Area of Orifice, A₀ = 5.06710⁻⁴ m²

Inner area of Pipe, A = 2.16 10⁻³ m²

Table 3: Calculated data for experimental volumetric flow rate, pressure head difference, square root of pressure head difference

Table 4: Calculated data for average velocity, theoretical volumetric flow rate, coefficient of orifice, Reynold's number

Appendix C

Sample Calculation for Observation No. 3

Density of water at 28, = 996.24 kg/m³

Viscosity of water at 28, = 0.0083238 Pa.s

Density of CCl₄ at 29.5, = 1575.71 kg/m³

Orifice diameter, D₀ = 1 inch = 0.0254 m

Pipe diameter (Schedule no. 40), D = 2.067 inch = 0.0525 m

Area of Orifice, A₀ = = 5.067 m²

Inner area of Pipe, A = = 2.16 m²

Volume of water after a half cycle of flow meter, V = 5.0 m³

Time interval for a half cycle of flow meter, t = 8.72 s

Experimental Volumetric Flow rate of water, Q_{Exp} = = 5.73 m³/s

Average Velocity of water, v = = 0.265 m/s

= 16.8 m H₂O

= 0.098 m H₂O

Square Root of Pressure Head Difference, = 0.312

Theoretical Volumetric Flow rate of water,

= 5.067 m³/s

= 7.21 m³/s

Coefficient of Orifice, C₀ = = = 0.795

Reynolds Number, Re = = =16644

Appendix D

Let section (1) be the upstream pipe section and section (2) be the downstream pipe section of orifice meter.

From the Continuity Equation,

Where,

Inner area of pipe Area of orifice,

Thus,

Applying Bernoulli's equation between section (1) and section (2),

Rearranging,

Let,

Then,

From continuity,

Substituting,

Velocity through the Orifice,

The theoretical volumetric flow rate,

And,

where, V = Volume of fluid t = Time interval

The theoretical discharge equation assumes ideal conditions with no energy losses. However, in real flow, losses occur due to viscosity, turbulence, flow separation, and vena contracta formation. Therefore, the actual discharge is always less than the theoretical discharge.

To account for these effects, the coefficient of orifice is introduced,

Hence,

For an ideal fluid, , but in practice its value is always less than unity.

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Obs. No	Experimental Volumetric Flow Rate, Q _{exp} (m ³ /s)	Pressure Head Difference, (m) ΔH	The Root of Pressure Head Difference, (m) ΔH ^{0.5}	Coefficient of Orifice, C ₀	Reynold's Number, Re
1	4.21	0.058	0.24	0.756	12216
2	5.18	0.081	0.28	0.789	15024
3	5.73	0.098	0.31	0.795	16644
4	5.79	0.112	0.33	0.751	16817
5	6.69	0.129	0.36	0.807	19429
6	6.75	0.151	0.39	0.752	19586
7	6.99	0.159	0.40	0.759	20298
8	7.41	0.175	0.42	0.767	21501

Symbol	Description	Unit (SI)
Q	Volumetric flow rate	m ³ /s
Q _{exp}	Experimental volumetric flow rate	m ³ /s
Q _{ther}	Theoretical volumetric flow rate	m ³ /s
ΔH	Pressure head difference across orifice	m
R	Manometer reading difference	m

Symbol	Description	Unit (SI)
A0	Area of orifice	m ²
A	Cross-sectional area of pipe	m ²
D0	Diameter of orifice	m
D	Diameter of pipe	m
v	Average velocity of fluid in pipe	m/s
C0	Coefficient of orifice	(dimensionless)
Re	Reynolds number	(dimensionless)
ρ	Density of fluid	kg/m ³
ρ _m	Density of manometer fluid (CCl ₄)	kg/m ³
μ	Dynamic viscosity of fluid	N·s/m ²
g	Acceleration due to gravity	m/s ²
V	Volume of collected fluid	m ³
t	Time interval for collection	s
P ₁	Pressure at upstream section	N/m ²
P ₂	Pressure at downstream section	N/m ²
z ₁	Elevation head at upstream section	m
z ₂	Elevation head at downstream section	m

No of Observation	Volume, V(L)	Time Interval, t(s)	Manometer Reading of CCl ₄ Left(cm)	Manometer Reading of CCl ₄ Right(cm)	Manometer Reading of CCl ₄ Difference(cm)
No of Observation	Volume, V(L)	Time Interval, t(s)	Left(cm)	Right(cm)	Difference(cm)
1	5.0	11.88	34.0	24.0	10
2	5.0	9.66	35.9	22.0	13.9
3	5.0	8.72	37.3	20.5	16.8
4	5.0	8.63	38.7	19.5	19.2
5	5.0	7.47	40.0	17.8	22.2
6	5.0	7.41	41.8	15.8	26.0
7	5.0	7.15	42.6	15.2	27.4
8	5.0	6.75	43.8	13.7	30.1

Observation No	Time Interval, t(sec)	Experimental Volumetric Flow Rate, Q _{exp} (m ³ /s)	Manometer Reading Difference, (cm)	Pressure Head Difference, (m)	Co-efficient Root of Pressure Head Difference, C _d
1	11.88	4.21	10.0	0.058	0.24
2	9.66	5.18	13.9	0.081	0.28
3	8.72	5.73	16.8	0.098	0.31
4	8.63	5.79	19.2	0.112	0.33
5	7.47	6.69	22.2	0.129	0.36
6	7.41	6.75	26.0	0.151	0.39
7	7.15	6.99	27.4	0.159	0.40
8	6.75	7.41	30.1	0.175	0.42

Observation No.	Average Velocity, v(m/s)	Theoretical Volumetric Flow Rate, Q _{theo} (m ³ /s)	Coefficient of Orifice, C ₀	Reynold's Number, Re
1	0.194	5.56	0.756	12216
2	0.239	6.56	0.789	15024
3	0.265	7.21	0.795	16644
4	0.268	7.71	0.751	16817
5	0.309	8.29	0.807	19429
6	0.312	8.97	0.752	19586
7	0.323	9.21	0.759	20298
8	0.342	9.65	0.767	21501

Assessment Component	Weight (%)	Obtained Marks
Summary	10%	
Introduction and Theory	15%	
Experimental Set up	15%	
Result and Discussion and Conclusion	25%	

Assessment Component	Weight (%)	Obtained Marks
Reference and Nomenclature	05%	
Quality of Tables	10%	
Quality of Figures	10%	
Overall Presentation	10%	
Total	100%	